

Math 526, S16

Name: Answer

Quiz 7

Section:

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix}$.

(a) Transform A into the reduced echelon form R .

$$A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = \text{rank}(A) = 2$$

Pivot variables: x_1, x_3

$$\text{Thus a basis for } C(A): \left\{ \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} \right\}$$

(c) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim(C(A^T)) = 2$$

$$\text{A basis for } C(A^T): \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 0 \end{bmatrix} \right\}$$

$$\text{or } \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the nullspace of A , $N(A)$.

$$\dim(N(A)) = 4 - \dim(C(A)) = 4 - 2 = 2$$

free variables: x_2, x_4 .

$$\text{Solve } RX = 0, \text{ i.e. } \begin{cases} x_1 + x_2 + \frac{1}{2}x_4 = 0 \\ x_3 + \frac{1}{2}x_4 = 0 \end{cases}$$

$$\text{i) } x_2 = 1, x_4 = 0 \Rightarrow s_1 = (-1, 1, 0, 0)$$

$$\text{ii) } x_2 = 0, x_4 = 1 \Rightarrow s_2 = (-\frac{1}{2}, 0, -\frac{1}{2}, 1)$$

Thus a basis for $N(A)$ is

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix} \right\}$$

(e) What is the dimension of the left nullspace of A , $N(A^T)$?

$$\dim(N(A^T)) = 3 - \dim(C(A)) = 3 - 2 = 1$$